

Survey paper

Adaptive and intelligent robot task planning for home service: A review

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ABSTRACT

The uncertainty and dynamic of home environment present great challenges to the task planning of service robots. The nature of the home environment is highly unstructured, with a diversity of tasks and a variety of objects. In addition, human, the service principle of robots, dynamically changes the environment and shows a large number of demands in the form of commands. Therefore, interaction is necessary. Robots must have the ability to adapt to complex scenes and act intelligently. Robot task planning for serving in unstructured and dynamic home scenes has been overlooked in the existing reviews. This paper firstly explores primary sources of uncertainty in-depth and summarizes three challenges, i.e., reliable planning under uncertain and incomplete information, efficient planning for complex tasks, and scalable planning for task generalization. Secondly, key solutions to the three challenges are discovered; the advantages and limitations of various task planning techniques are assessed from the perspective of adaptability and intelligence. Finally, the promising avenues toward overcoming uncertainty, improving satisfaction and comfort of users, and enhancing learning skills and generalization are proposed.

1. Introduction

What would it be like to have a robot in your home life? In the movie “Robot and Frank”, there is such a robot, which has powerful artificial intelligence, can talk to anyone in fluent language, can cook, clean the house, check the physical condition of old Frank, and even cultivate old Frank’s healthy habits. It sounds wonderful, but as technology advances, it may no longer be something in science fiction movies.

The reality is that robots are gradually entering human homes to provide service, becoming new types of smart equipment and even good partners for humans to improve their quality of life.

Task planning of home service robots generates sequences of actions to complete the given tasks, which is based on high-level actions similar to the semantic abstractness of human language such as grasping, picking, and placing, and can reflect the action process (Zhang et al., 2021). In addition, low-level actions that are parameters of actions such as joint angle, motion speed, and approach direction also need to be planned in detail, because it is the bridge connecting task planning and execution (Wang and Tian, 2022). In this paper, the generation of high-level actions is focused on.

Unlike classical planning hypotheses of determinism, full observability, the robot’s sole agent of change, and reachability goals (Ghallab et al., 2004), task planning in the home environment is fraught with uncertainty and changes (Pajarinen and Kyrki, 2017; Wang and Tian, 2022) and requires consideration of human–robot interaction (HRI) (Ji

et al., 2020). Meanwhile, manipulation skill is also an important topic. All of the above require robots to be more adaptive and intelligent. Therefore, task planning for home service is an inherently challenging and crucial issue.

Contemporarily, most task planning methods (Nasir et al., 2019; Savage et al., 2019) address problems in a deterministic environment, which can generate a static sequence of task execution adaptive for certainty rather than uncertainty. By contrast, adaptability and intelligence are innate to humans.

Humans have excellent task planning skills, even in an unknown home environment. For example, opening medicine bottles (Edmonds et al., 2019), which may require pushing the cap to unlock (visually invisible), unscrewing it, then pulling it open, is a challenging multistep manipulation for robots but easy for humans in any home scene. Human ability of manipulation is not easily disrupted by changes in the environment as different policies to find the target objects relevant to tasks are employed, which is of great adaptation (Wang et al., 2020). Moreover, once a person knows how to open medicine bottles, it will be easy to open things that have never been encountered before, such as glue sticks. These explorations and actions raised are the essence of true intelligence (Seligman et al., 2013).

How long does it take for robots to work appropriately and productively at home? At present, in unstructured environments, robots inevitably face uncertain and incomplete information while adhering to the principle of being people-oriented. Therefore, how to obtain enough home environmental information and to consider the naturalness of

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2.3.1. Reliable planning under uncertain and incomplete information

To act intelligently, robots must be able to deliberately plan for what they will act. However, in a cluttered and unstructured environment, objects may be partially observable or dynamic, and noise may exist. Determining how to search for a target object and enable robots to reason about perceptual and manipulative uncertainty during the search is of great importance. Moreover, the state of the home scene is likely to change due to the execution of actions and human movement, which will result in task failure. Determining how to make robots adapt to uncertainty and changes as well as re-plan to finish given tasks is a key issue.

2.3.2. Efficient planning for complex tasks

In many real-world environments, plans must be generated quickly, both at the beginning of execution and after each execution failure (Martinez et al., 2016). In the meantime, timeliness is critical for achieving natural and smooth interactions, especially when robots not only interact with people but also with the environment through sensors such as vision, producing vast quantities of information. Moreover, the diversity of service tasks and the uncertainty of target location increase the complexity of task planning, especially in the unstructured and dynamic home scene. In this case, determining how to simplify the complexity of tasks, raise efficiency, and even re-plan is another challenge.

2.3.3. Scalable planning for task generalization

Planning for each task is unrealistic due to the unstructured nature of the home environment and the diversity of objects. Therefore, determining how to find the identity of similar tasks and realize generalization performance is a topic of great value. For example, when a robot learns to twist a medicine bottle, it should be able to twist a glue stick.

3. Probabilistic planning under uncertain and incomplete information

The state of the real home environment is incomplete and uncertain because of clutter and occlusion. The probabilistic planner, focusing on the uncertainty of action or reasoning, has been applied to search for a target object and generate sequences of actions. Kollar and Roy (2009) developed a probabilistic model to search for unacquainted objects by utilizing possible object locations in object-object and object-scene contexts. Song et al. (2015) presented a probabilistic framework for representing and modeling grasping tasks with the ability to generalize tasks and objects. In addition, probabilistic reasoning can handle ambiguous task commands by letting the robot know where the target object is Wang and Tian (2022). However, probabilistic planning is typically performed offline.

3.1. POMDP+ online

There is no doubt that offline planning is suitable for deterministic environments with known and unchanged conditions. However, failure may arise when the environment changes. In this case, online planning with the characteristics of interaction with the environment enables task planning in partially observable and dynamic domains. These problems can be described as a partially observable Markov decision process (POMDP) (Kaelbling et al., 1998). POMDP can model the uncertainty of observations and action outcomes. The problem of a POMDP is decomposed into two parts, as shown in Fig. 2.

Both Markov decision process (MDP) and POMDP agents perform actions and receive immediate rewards. However, MDP agents assume direct observation of the state of the environment; while the POMDP agents perceive an observation generated by the perceptual process. A POMDP agent can be mainly divided into two parts: (i) State estimator: it uses the latest observation, action, and state to generate the updated

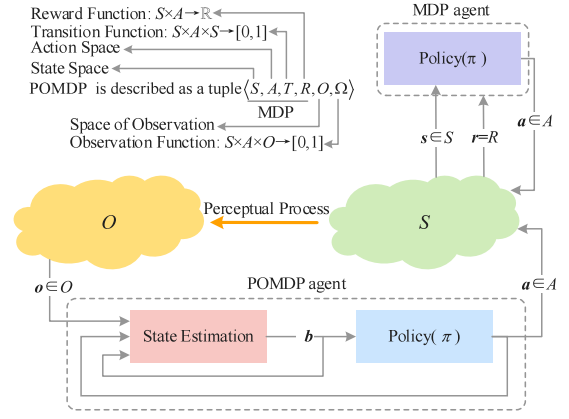


Fig. 2. Schematic description of POMDP.

state of belief (**b**). (ii) A policy (π): it inputs the current belief state, outputs the action to be taken, and takes the optimal step in the next action to maximize the reward R .

As shown in Fig. 2, a POMDP can be defined as a $\langle S, A, T, R, O, \Omega \rangle$ (Zhao and Chen, 2021). At each step, the agent executes an action a under state s ; changes the states to s' with the probability $T(s, a, s') = \Pr(s' | s, a)$; obtains an immediate reward $r = R(s, a)$; receives a new observation o with the probability $\Omega(s', a, o) = \Pr(o | s', a)$.

In the partially observable environment, the sensor does not observe a particular world state but a probability distribution over the set of world states. The agent needs to hold a belief state $b \in B$, where B is the belief space containing all possible belief states. Denoting the probability of assigning a belief state b to an actual state s as $b(s)$. Starting from the initial belief state b_0 , the state estimator must update $b(s)$ at each time step after the agent takes action a and observes o . The belief update equation can be obtained from the Bayes formula

$$b'(s') = \alpha \Pr(o | s', a) \sum_{s \in S} \Pr(s' | s, a) b(s) \quad (1)$$

where $\alpha = 1 / \Pr(o | a, b)$ is a normalizing constant that makes $\sum_{s \in S} b'(s') = 1$

A policy is a mapping $\pi: b \rightarrow a$, its value function $V_\pi(b)$ represents the expected total discounted reward under the policy π at the current belief state b :

$$V_\pi(b) = E \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, \pi(b_t)) \right] \Big|_{b_0=b} \quad (2)$$

where $\gamma \in (0, 1)$ is a discount factor that determines the impact of future rewards on current decisions.

The goal of a POMDP planning is to find an optimal policy π^* , which maximizes $V_\pi(b)$ over B . According to the Bellman equation, the optimal value of a belief state is defined as

$$V^*(b) = \max_{a \in A} \left[\sum_{s \in S} b(s) R(s, a) + \gamma \sum_{o' \in O} \Pr(o' | b, a) V^*(b') \right] \quad (3)$$

While as more objects are considered, the dual challenges of dimensional curses on intricate models of the real world and historical curses on complex reasoning about objects and their relations grow exponentially. With offline methods, the algorithm returns a policy defining which action to execute in each possible belief state. Such approaches tend to work only when dealing with small to mid-size domains because policy generation takes significant time. On the contrary, online-based methods perform tasks by alternating calculations and actions with meeting short real-time constraints. In this case, the time available to generate a plan is often substantially shorter. Therefore, the POMDP+ Online method can handle more complex task classes as it can reduce the complexity of the state space given to the POMDP solver

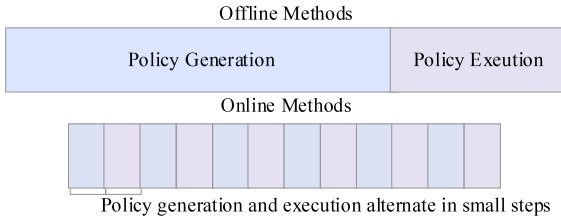


Fig. 3. Comparison between offline and online methods.

(Grady et al., 2015), and many improvements have been achieved. The difference between offline and online methods is shown in Fig. 3.

Pajarinen and Kyrki (2017) introduced particle filtering and occlusion ratio to online POMDP planning, which can produce compact policies to deal with the dynamics of the world model and performance constraints. Zhang et al. (2017) introduced interleaved commonsense reasoning and probabilistic planning (iCORPP) to construct POMDPs for adaptive robot planning dynamically. Yang et al. (2021) incorporated an online POMDP planner into the Behavior Trees (BTs) reactive executive for interleaving planning. Adu-Bredu et al. (2021) and Garrett et al. (2020) designed online planning and execution system to do housework in the kitchen. Although replanning is designed to deal with contingencies, its success rate in practice has not been satisfactory, especially in the case of high real-time requirements.

To ensure that the task is done correctly, Wang et al. (2021a) combined policy with safe-reachability objectives and a partial conditional plan to ensure a bounded replanning probability. However, it cannot deal with continuous workspace directly. Online algorithms with continuous state, action, and observation spaces (Jiang et al., 2019; Sunberg et al., 2018) can guide to solve the problem. Nouman et al. (2021) introduced a parallel algorithm to yield a new offline hybrid conditional planner (HCPLAN) to avoid many replanning during the execution but the method requires more time to compute. At this point, it is of great importance to introduce online planning and to consider planning for the most likely contingencies rather than the probable ones.

Meanwhile, Determinized Sparse Partially Observable Tree (DESPOT) (Somani et al., 2013) as an approximate online POMDP algorithm that can handle large state spaces has been used widely to greatly improved the planning efficiency of online POMDP under uncertain conditions and made it a valuable tool in robotics practice (Cai et al., 2021; Luo et al., 2019). Furthermore, Zhao and Chen (2021) proposed a hierarchical POMDP (HP) planning framework. As

Table 1

Experiment results (Zhao and Chen, 2021).

		Success	Number of steps	Execution time (s)
(a)	HP	✓	4	57.97
	NP	✓	4	59.73
	SP	✓	6	95.75
(b)	HP	✓	4	56.87
	NP	✓	6	83.96
	SP	✓	10	157.86
(c)	HP	✓	6	80.71
	NP	✓	12	146.76
	SP	✓	10	157
(d)	HP	✓	6	82.42
	NP	✓	10	126.7
	SP	✓	10	157.31
(e)	HP	✓	8	106.42
	NP	✓	12	147.26
	SP	✓	14	208.52
(f)	HP	✓	4	57.72
	NP	✓	4	57.38
	SP	✓	14	222.56

illustrated in Fig. 4, the framework consists of hierarchical POMDP modeling and planning. It updates unknown probabilities of transition and observation functions during task execution.

Experiments for the object fetching task have been conducted. 6 typical scenes are selected. The scenes for experiments are shown in Fig. 5, in which the red can is the target. Tasks are executed with HP, Normal POMDP (NP), and Systematic policy (SP) planning methods. Results of the experiment are shown in Table 1.

As shown in Table 1, all the 18 tasks are successful, but HP always has the best performance and SP usually spends the most steps and execution time, which shows that the method is effective and efficient. Moreover, the method is robust because the results of experiments are conformable to the simulations while the objects, shelf, and observation model are different from the simulation. All of these make it convenient to tune parameters and learn probabilities of the abstract POMDP until it can adapt to real tasks in simulations, then migrate them to a physical system. Although objects are different, they are still cylinders. The manipulator mounted with a Microsoft Kinect will return to a predefined pose after each action to take an observation. Therefore, object complexity and flexibility of manipulation should be taken into consideration.

Table 2 compares various methods proposed for task planning under uncertainty with their performance, experiments, limitations, and

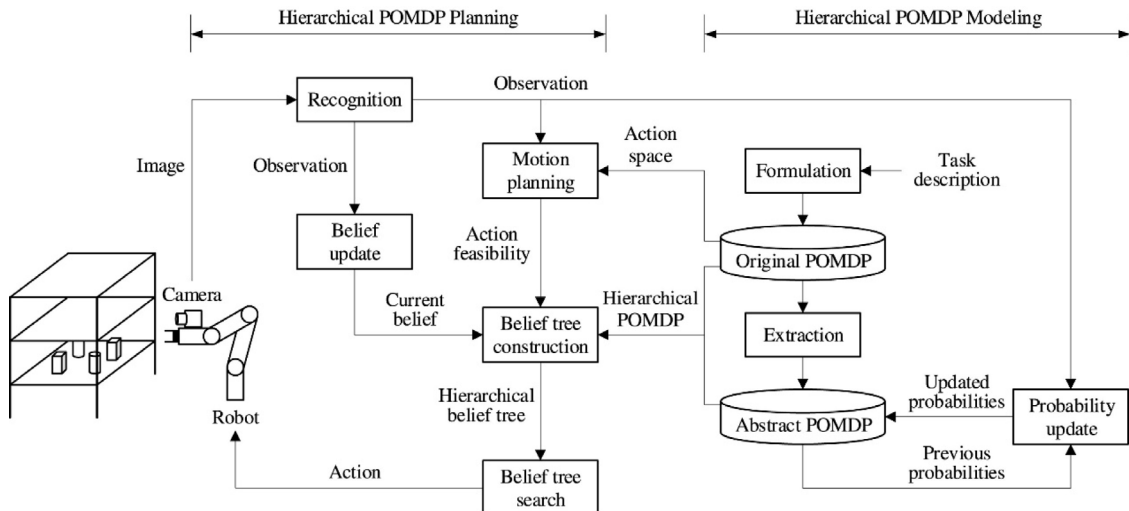


Fig. 4. Hierarchical POMDP planning framework (Zhao and Chen, 2021).

Table 2

Comparison of various methods for task planning under uncertainty.

Ref.	Dynamic	Mobile robots	Obstacle avoidance	Objective	Method(s) /algorithms	Accuracy /performance	Experiments	Limitation/possible improvements
Pajarinen and Kyrki (2017)	✓		✓	Manipulation of multiple unknown objects in crowd	Particle filtering, occlusion ratio, and grasp probability	Offline and online multi-step planning	Finding and moving dirty dishes into a dishwasher	Prior knowledge and learning for autonomy
Zhang et al. (2017)	✓	✓	✓	Dynamically reasoning about	Declarative language P-LOG and iCORPP	Adapting to the changes	Shopping request identification, robot navigation	Decreasing reasoning time
Yang et al. (2021)	✓	✓	✓	Handling both partial observability and unexpected dynamics	Interleaving task planning, BTs	Reactive in run-time response and efficient	Chase-and-hide task	No manipulation
Wang et al. (2021a)	✓	✓	✓	Safe-Reachability	Online Partial Conditional Plan Synthesis	Greatly improving the computational efficiency	Picking up the can on the table while avoiding collisions	Dealing with continuous workspace directly
Garrett et al. (2020)	✓	✓	✓	Multistep manipulation and reacting to unexpected	Online planning and execution system	Reduction in total planning time	Inspecting and swapping drawers, cooking spam tasks	Improvement in the success rate of swapping and cooking
Nouman et al. (2021)	✓	✓	✓	Consider contingencies while planning	Offline hybrid conditional planner	Avoiding many re-planning	Setting up a table for lunch	Needing more time and incomplete knowledge
Luo et al. (2019)	✓	✓	✓	Online planning under uncertainty	Introducing importance sampling, DESPOT	Significantly improves the performance	Single-seater autonomous vehicle	Not fully automated, needing an iterative learning scheme
Zhao and Chen (2021)			✓	Solving the constraints and uncertainties of object manipulation	Hierarchical belief tree search	High average performance in repeatedly task	Picking or placing objects in clutter	A limit to the number of actions

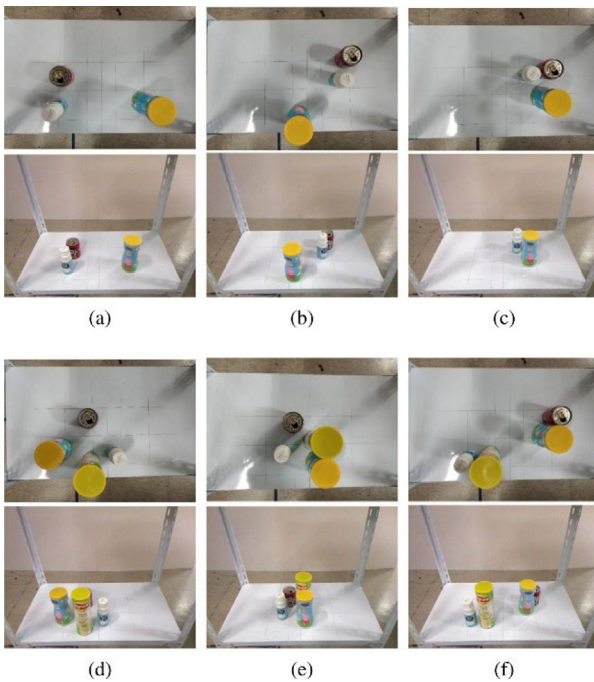


Fig. 5. Scenes for experiments. The up picture of each scene is the top view and the bottom is the robot's view. In scenes (c) and (d) the target is fully occluded; in scenes (b) and (f) the target is occluded a lot; in scenes (a) and (e) the target is occluded a little (Zhao and Chen, 2021).

possible improvements, illustrating that knowledge and learning apart from computation improvement are significant to achieve good performance.

3.2. POMDP+ retrieval manipulation

Manipulating objects is one of the most significant skills for robots to serve in the home environment, which requires the robots to bring objects to a user from a cluttered, or even high-occluded scene, such

as retrieving a target from a tabletop, a fridge, or a shelf. In this case, removing some objects at the front with incomplete perception and no collision is challenging but necessary. Although manipulation under uncertainty is inevitable, uncertainty can be reduced or eliminated during the process.

There are some methods proposed from different perspectives to retrieve occluded objects. Li et al. (2016) used the DESPOT; Nie et al. (2016) employed Markov-chain Monte Carlo (MCMC), and space constraints were taken into account to improve efficiency, but there is still a long way to go for human assistance in complex domains. Xiao et al. (2019) proposed Parameterized Action Partially Observable Monte-Carlo Planning (PA-POMCP) to solve the difficulty of manipulation, which evaluated manipulative actions by considering the impact of the robot's current beliefs on the success of action execution. Lee et al. (2019) employed a collision avoidance scheme that had been widely used in mobile robot navigation to adapt to dynamic changes in the configuration and introduced a histogram to calculate the grab path. Ahn et al. (2021) and Nam et al. (2021) utilized traversability mapping to represent the target and the object to be relocated as nodes for finding the min-hop path while outfitting the relocation planner to deal with failure.

Amiri et al. (2022) proposed scene analysis for robot planning (SARP) to reason with contextual information and used mobile navigation technology for searching for a target object in unknown large-scale environments. However, it assumed that humans are stationary and thus both perception and reasoning capabilities should be improved. Kim and Sentis (2022) integrated Dynamic Bayesian Networks (DBN) model, particle filtering, and POMDP planning for active object tracking under various dynamic conditions. The situation of finding a target when it suddenly disappears from the robot's field of view (FOV) and is not believed to be occluded by another object is showcased in Fig. 6. The time analysis for an example scenario of active object tracking is illustrated in Fig. 7.

The action sequence shown in Fig. 7 demonstrates the capability of robot to actively find occluded or missing targets under various dynamic conditions. As shown in Fig. 8, the rate represents the success rate of tracking or finding a target after the operation is executed, based on the state of the context that is believed. Track has the highest success rate of 0.88, followed by active moves, while search has the highest variance and lowest success rate of 0.74. In addition, four criteria

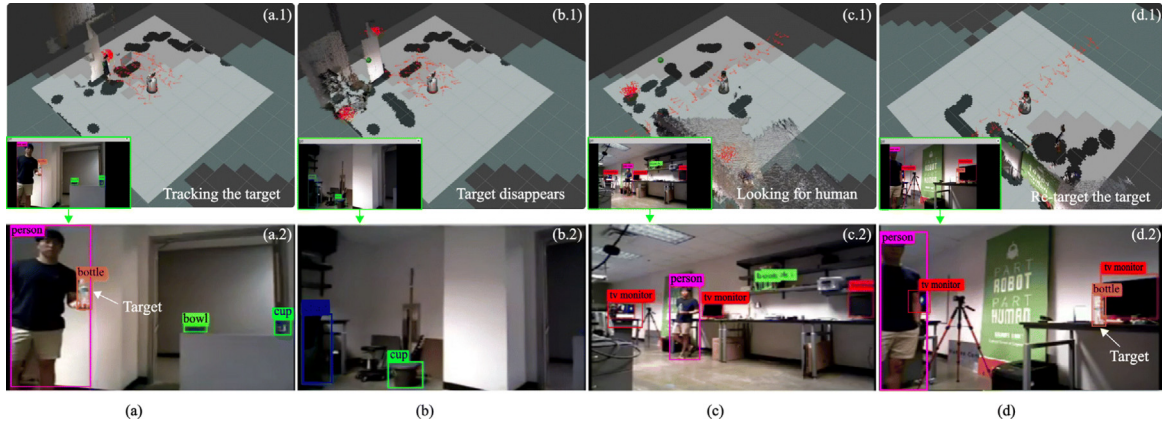


Fig. 6. Finding a target when it suddenly disappears from the robot's FOV and is not believed to be occluded by another object. (a) Initially, the robot tracks the target (a bottle) using its depth cameras. (b) The target suddenly disappears from its FOV; here it is supposed that the target has disappeared because of the information returned from the observation model. (c) The robot moves to find a person around its neighborhood. If a person is detected, the robot navigates to that person and attempts to locate the target nearby. (d) The robot succeeds in re-locating the object since it is placed next to the person (Kim and Sentis, 2022).

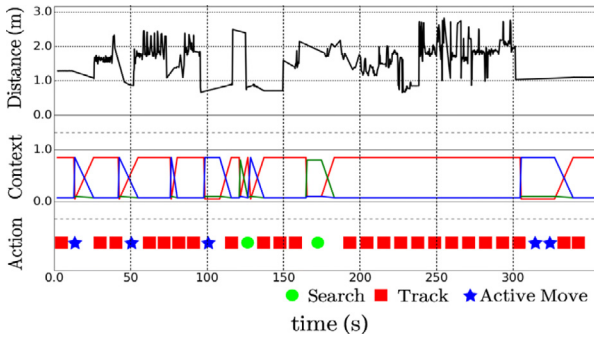


Fig. 7. Time analysis for an example scenario of active object tracking. The first row shows the estimated distance between the robot and the target object. The second row displays the estimated context state (red line: visible, blue line: occluded, and green line: disappearance). The third row provides the chosen actions at each time frame (Kim and Sentis, 2022).

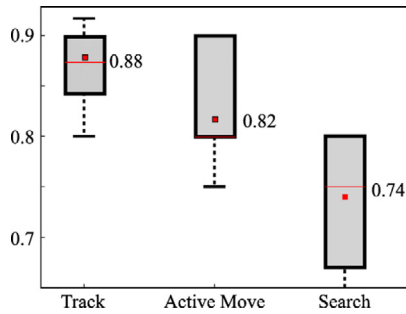


Fig. 8. Success ratio for various active sensing scenarios (Kim and Sentis, 2022).

Success Ratio (SR), Tracking Ratio (TR), Average Restoring Time (ART), and Failure Time (FT) are applied to verify robustness and efficiency, the mean values are 0.82, 0.71, 10.22 s, 232 s, respectively. In a word, the ability that robots can perform practical tracking and searching tasks in dynamic environments is verified convincingly. However, it also has a few unrealistic assumptions and a memory-based approach is desired to predict context.

Moreover, visual servo (Donmez and Kocamaz, 2020; Donmez et al., 2018) made mobile robots have high accuracy and efficiency. Further, Liu et al. (2017) and Zou et al. (2021) introduced biology-inspired episodic memory to empower robots with experience and cognitive ability to implement common tasks in a complex environment with

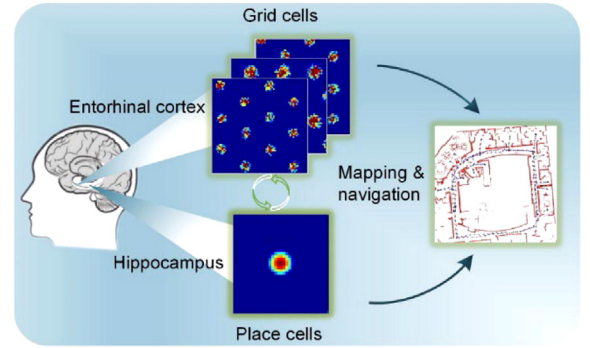


Fig. 9. The schematic of the brain positioning system (Zou et al., 2021).

obstacles. The simplified schematic of the brain positioning system is illustrated in Fig. 9, in which place cells build a map of their internal environment; grid cells provide a spatial metric coordination system.

Table 3 compares various methods proposed for retrieval manipulation under uncertainty with their performance, experiments, limitations, and possible improvements. It reveals that these methods focus on either retrieving the target object in clutter or guiding robots to move to the target accurately. Nevertheless, fulfilling more real-world tasks, such as receiving an instruction to make a cup of tea, which involves different locations, requires dual-arm manipulation and multiple steps. Moreover, large-scale and high dimensions of the real environment are in urgent demand of robots with cognitive and reasoning abilities, thus robots can provide comfortable service for people with efficient and changeable working modes in the home environment.

3.3. Knowledge-based approach

The knowledge-based approach is another promising way to handle uncertainty for the reason that the more robots acquire, the fewer unknowns they will encounter.

Prior knowledge has been used to assist robots in task planning. Lorbach et al. (2014) proposed a priori-assisted propagation of spatial information to search for objects. Zhang et al. (2019) introduced a dynamic search method for target objects based on prior knowledge, which was inspired by the human search process to model the common knowledge of the typical spatial locations and the relation among objects as prior knowledge. Then, prior knowledge was employed to provide the probability models of action-object correlation, object cooccurrence, active collaboration, and constructed through statistical

Table 3

Comparison of various methods for retrieval manipulation under uncertainty.

Ref.	Dynamic	Mobile robots	Obstacle avoidance	Objective	Method(s) /algorithms	Accuracy /performance	Experiments	Limitation/possible improvements
Xiao et al. (2019)		✓	✓	Target object search in clutter under uncertainty	PA-POMCP	Solving the target object search task in the settings	Retrieving a stapler from a tabletop	Moving to get a good view
Lee et al. (2019)	✓		✓	Obstacle rearrangement in clutter	A collision avoidance scheme for mobile robot navigation	Improving up to 31% of the execution time	Taking out a target from a shelf or a fridge	Dealing with the uncertainties regarding the poses of the object
Nam et al. (2021)	✓	✓	✓	Object Retrieval Manipulation in clutter	Traversability graph and graph structure	At least 28.0% of reduction in a known static environment with 20 objects	Optimizing the number of pick-and-place actions	Assumptions and more underlying requirements
Ahn et al. (2021)	✓		✓	Object Retrieval Manipulation in clutter	Traversability graph and graph structure	Reducing planning time up to 74.9% and higher success rate	Retrieving the target object in the back	Extending to the mobile robot and bulky end-effector
Kim and Suh (2019)		✓	✓	Active object search in the unknown and large-scale room	Probabilistic environment model with knowledge and spatial relations	Better than the coverage-based and random search methods	Search objects in the house consisted of 7 room categories	Quicker target object search in familiar environments
Amiri et al. (2022)	✓	✓	✓	Achieving long-term goals under uncertainty	Scene analysis for robot planning	Improving both efficiency and accuracy	Searching for a banana navigates	Assumptions, improvement of both perception and reasoning
Zou et al. (2021)		✓	✓	Improving the cognitive ability in unknown	Episodic memory based self-organizing learning	Adapting to the changes in the environment	Guiding robots to move to the target accurately	Needing experience and semantic representation
Kim and Sentis (2022)	✓	✓	✓	Active object tracking under dynamic state	DBN model, particle filtering and POMDP	Relocating targets with high success rates	Object tracking	Assumptions, using memory to predict context, RNN

learning (Zhang et al., 2021). Meanwhile, the ontology technique, as a knowledge-based representation technique, can provide information data related to the robot world to enhance task planning performance. Relevant studies have been published, such as presenting a robot knowledge framework to improve intelligence (Lim et al., 2011; Suh et al., 2007) or providing service of HRI (Chang et al., 2020). More information about knowledge representation in service robots can be found in Paulius and Sun (2019).

Earlier studies have shown that using semantic knowledge in task planning makes robots more capable of solving tasks without human intervention, handling situations that are not explicitly considered in the design and planning in large domains (Galindo et al., 2008b). Some scholars have tried to promote the application of semantic knowledge to robot task planning to realize reasoning (Aydemir et al., 2013; Cui et al., 2021; Savage et al., 2019). Moreover, Tenorth and Beetz (2017) proposed the open-source software KnowRob for providing robots with comprehensive knowledge and inference capabilities. The process of integrating external inference machines into the web ontology language (OWL) model that serves as a conceptual framework for storing the information they produce and consume is shown in Fig. 10. Three example applications of KnowRob on physical robots i.e. PR2 (left) and the Amigo robot (center) serve a drink from a cupboard in different environments as well as the pancake-making task (right) are illustrated in Fig. 11.

Furthermore, Beetz et al. (2018) presented KnowRob 2.0, an extension and partial redesign of the former KnowRob (Tenorth and Beetz, 2013) that substantially improved the abilities of robots to acquire open-ended manipulation skills and common knowledge as well as execute actions more realistically. A comparison of different knowledge-based methods with advantages and limitations is summarized in Table 4.

Besides, some studies focused on integrating knowledge and probability for searching target objects efficiently to generate a better sequence of actions (Liu et al., 2022; Wang et al., 2020). Wang et al. (2020) discovered the co-occurrence relation between dynamic objects

and static objects (as shown in Fig. 12) and constructed a probability model of dynamic and static objects' location relations to determine the location of objects.

$$\rho(o_i | o_j) = \frac{N(o_i, o_j) + \tau}{N(o_j) + \tau n} \quad (4)$$

where o_i and o_j represent dynamic objects and static objects, respectively. $N(o_i, o_j)$ represents the number of times o_j , and $N(o_i, o_j)$ represents the number of times objects o_i and o_j . n denotes the number of detected objects and τ plays a smoothing role.

If the target dynamic object o_T is observed, the probability of finding o_j can be obtained through Bayes's law:

$$\rho(o_j | o_T) = \frac{\rho(o_T | o_j) \rho(o_j)}{\sum_{o_j \in \theta} \rho(o_T | o_j) \rho(o_j)} \quad (5)$$

where θ denotes a set, which represents objects detected. The summation on j represents the sum of all members of θ .

The probability of successfully finding the target object based on the static object O^* related to the target dynamic object is maximized according to Eqs. (4), (5).

$$O^* = \arg \max_{o_j \in \theta} [\rho(o_j | o_T)] \quad (6)$$

The overall knowledge-based framework of object search for robot service planning that includes the knowledge layer, the planning layer, and the service layer is delineated in Fig. 13

Task re-planning strategies that consider changes in target position were designed to improve the adaptability of robots in the real world (Wang and Tian, 2022; Wang et al., 2020). The overall framework of the hybrid task planning approach (Wang and Tian, 2022) is shown in Fig. 14.

To verify the effectiveness of the method, an experiment with the Tiago robot to execute the task of transporting a box of milk is demonstrated in Fig. 15 and the sequences of actions are: (a) move

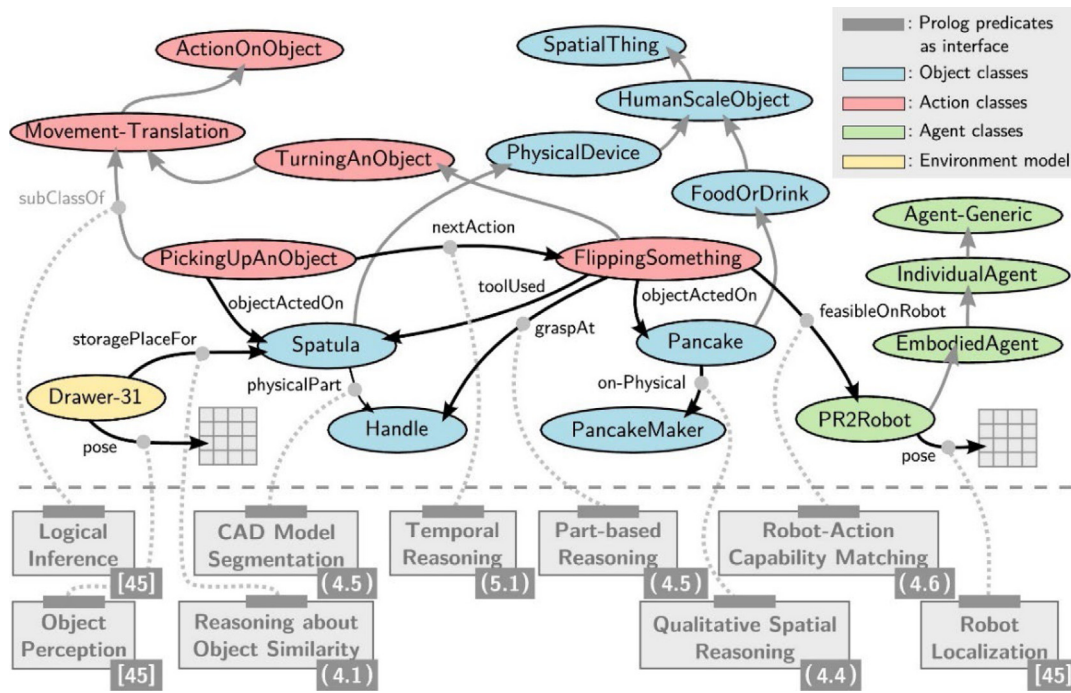


Fig. 10. The process of integrating external inference machines into the OWL model (Tenorth and Beetz, 2017).



Fig. 11. Three example applications of KnowRob on physical robots (Tenorth and Beetz, 2017).

Table 4

Comparison of different kinds of knowledge-based methods.

Type	Definition	Function	Advantages	Limitations
Prior knowledge	Common/general/default knowledge	Inferring the locations of the target objects and dynamic object search	Reducing search space and efficient object search	Incomplete and Update to maintain its reliability
Ontology knowledge	Concepts and relations for knowledge sharing across multiple domains	Representing the knowledge and model of robots	Shareable and Exchangeable	Conflicts or misconceptions may arise due to inconsistency
Semantic knowledge	Positions of ontology and positioning relation between dynamic and static objects	Offering environmental information and inferring the location of the target object	Efficient, reliability and effectiveness	Target objects may not be found when acting and re-planning may be needed
KnowRob	Knowledge base (KB)	Providing knowledge for performing everyday manipulation tasks	Reducing the time for constructing large KB	Difficult to perform complex manipulation tasks
KnowRob 2.0	KB	Acquiring commonsense knowledge, open-ended manipulation skills, and competence	Performing complex manipulation tasks	Complexity system

to the kitchen, (b) approach the refrigerator, (c) open the door of the refrigerator, (d) close the door of the refrigerator, (e) move to the dining room, (f) approach the refrigerator, (g) use POMDP to generate online action sequences, (h) move to the living room, (i) approach the tea table, (j) place the box of milk on the tea table.

As seen in Fig. 15, a hybrid task planning method can decompose the complex task into an offline and online combination of action sequences and adapt to the dynamic changing home environment. However, the relations between actions and objects in the primitive task are still mediated manually without learning ability. Enabling

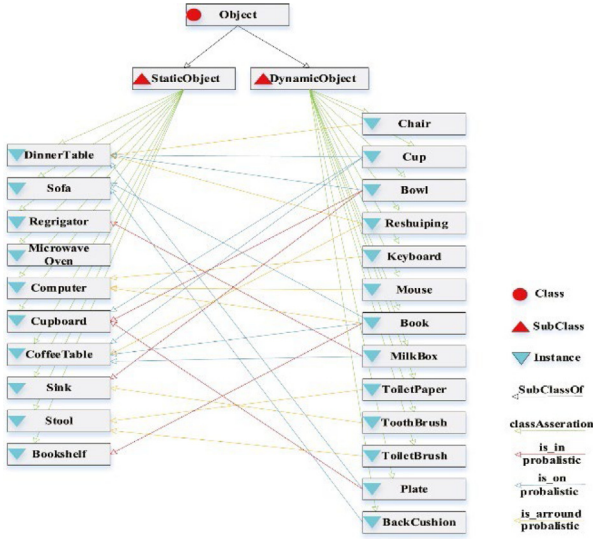


Fig. 12. Location relation between dynamic and static objects (Wang et al., 2020).

a robot to know well the decomposition of complex tasks and the relations between actions and objects through reinforcement learning is urgent to be solved.

In addition, commonsense knowledge is also an available way to reason and plan under uncertainty for active and adaptive robot planning dynamically (Kim and Suh, 2019; Zhang et al., 2017). Although the interaction with the environment or humans will increase the difficulty of task planning, the knowledge of robots can be enriched, which is also an exciting means to enhance the ability to handle incomplete information and unexpected changes (Chen et al., 2016; Gomez and Miura, 2021; Lim et al., 2014). These approaches are useful for task planning in uncertain environments, however, more efforts should be made to diagnose task planning execution, in addition, to sift through and process the vast amount of information generated by HRI.

4. Hierarchical Task Network for complex tasks

The diversity of service tasks and the variety of objects in the unstructured and dynamic home environment increase the complexity and difficulty of task planning. In the field of task planning, there are many approaches to reason about actions and plans, whereas Hierarchical Task Network (HTN) is beneficial to solving real-world planning problems as it can better characterize the way that users ponder problems (Balaguer et al., 2016; Tomov et al., 2020). In 1975, Nets

of Action Hierarchies (NOAH) (Sacerdoti, 1975) proposed the original idea of hierarchical planning. Subsequently, various HTN planners have been developed, the most famous of them are the simple hierarchical ordered planner (SHOP) and its successor, SHOP2 (Nau et al., 2003, 2005, 1999). These planners make HTN planning more efficient due to the powerful expressiveness of predefined domain knowledge, which can recursively break down an abstract task into smaller and smaller subtasks until each subtask reaches an action (Nau et al., 2003, 1999). The schema of HTN approach is shown in Fig. 16.

An HTN planning problem Π can be defined as a tuple $\langle s_0, \omega_0, O, M, Q \rangle$, in which methods M are used to recursively break down tasks into smaller and smaller subtasks until the planner reaches a primitive; then indeterminately chooses a task node and a method is represented by two dotted rectangles. If a task is nonprimitive, the planner indeterminately selects an applicable method and instantiates it to break the task into subtasks. If not, the planner selects an applicable operator and instantiates it to generate an action. Only when all the constraints are satisfied does the planner find a solution; otherwise, backtracking and trying other methods or other instantiations are needed.

4.1. HTN for HRI with user preference

Due to the ability to express complex problem-solving knowledge, HTN can solve the two major demands of HRI to process large amounts of information and high efficiency, which is the highlight of home service. Galindo et al. (2008a) proposed a symbolic model of the environment in a natural form of structuring knowledge with multiple hierarchies through a human-inspired mechanism to enable the HRI task planning available in the real world, where hierarchies are used to improve the efficiency of the algorithm and communicate with a human, respectively. Multi-user applications are allowed under human preference. Therefore, user preference is significant in the HRI field and had been used vastly to quickly generate high-quality plans.

Li et al. (2009) focused on learning probabilistic hierarchical task networks (pHTNs) that capture the preference of users by the means of checking user-generated plans without prior information. An algorithm to automatically learn pHTNs via observing what users act was introduced (Li et al., 2014). Das et al. (2019) proposed the PGPLANNER framework to improve the relevance of preference since planners actively solicit as needed, and uncertainty in plans can be computed explicitly. The overall architecture of the PGPlanner framework is shown in Fig. 17. However, this approach does not validate preferences and assumes that the user is an expert, so it is difficult to obtain an optimal solution that satisfies the user's preferences.

Shao et al. (2021) firstly introduced Monte Carlo Tree Search (MCTS) in HTN planning to propose the MCTS-HTN algorithm and then

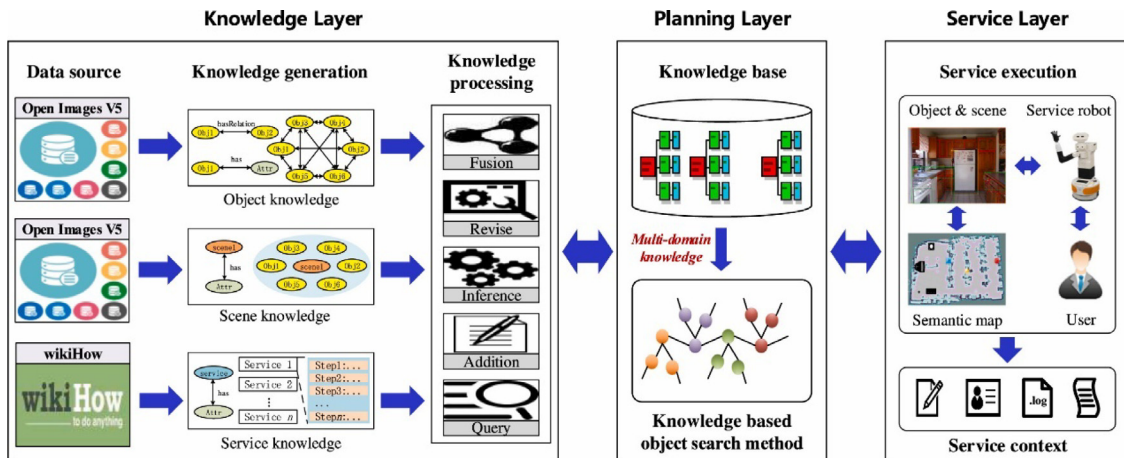


Fig. 13. The overall knowledge-based framework of object search for robot service planning (Liu et al., 2022).

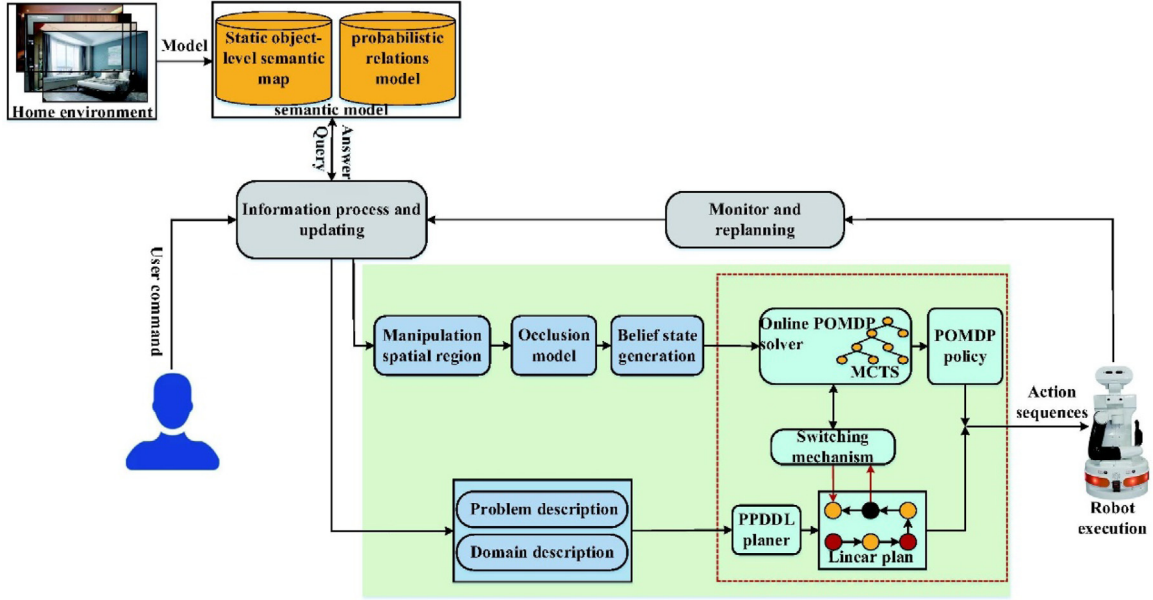


Fig. 14. Overall diagram of hybrid task planning for a service robot (Wang and Tian, 2022).

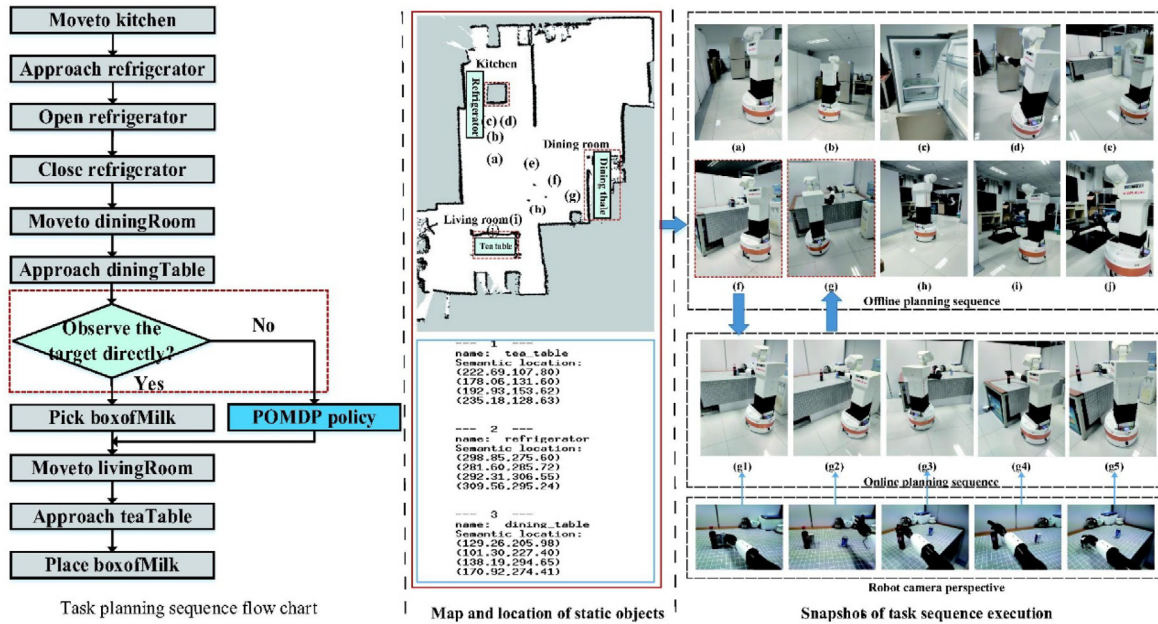


Fig. 15. An example of milk delivery task (Wang and Tian, 2022).

integrated it with the SHOP planning algorithm to construct the SHOP-m algorithm, which is also named Pyhop-m as it is implemented in Python. SHOP-m can reduce the influence of domain knowledge on planning results and obtain real-time optimal solutions, regardless of whether the environment is static or dynamic. The specific planning and execution process of SHOP-m in a dynamic environment is shown in Fig. 18.

In practice, changes in the environment will cause the premise of the original task to be unsatisfied, thus it is necessary to replan for completing the remaining tasks in the current environment. Planning result in a dynamic environment is shown in Table 5.

As seen from Table 5, the Pyhop-m algorithm can obtain a smaller plan length with higher stability at each set of experimental points. By calculating the confidence intervals of four probabilities at 10 points, the confidence intervals of the planning success rate and the optimal solution rate of the Pyhop-m algorithm are [75.82%,89.18%] and

[88.67%,93.95%], respectively. While the confidence intervals of the baseline are [58.19%,77.81%] and [69.91%,80.69%], which indicate that the Pyhop-m algorithm also has a higher and more stable success rate and the optimal rate in the dynamic environment.

However, the optimal scheme is obtained under the condition that the external environment is fully observable. Moreover, since the replanning of Pyhop-m comes from utilizing MCTS, determining when to replan and how to make good use of the planning tree of the last plan are all hot spots for future study on this method.

Motivated by increased trust between users and agents, Lyu et al. (2021) proposed a unified framework of Trustworthy Decision-Making for the planning of complex and higher-level composite tasks by reasoning about explicitly expressed knowledge and learning interpretable subtasks based on symbolic planning. Bougie and Ichise (2022) achieved a dramatic acceleration of the convergence of sub-strategies to reach any sub-goals by introducing preference from non-expert humans

Table 5
Planning results in a dynamic environment (Shao et al., 2021).

Initial	Target	Step_h	Step_m	P1_h	P1_m	P2_h	P2_m
1,0	2,6	[7.93,8.66]	[7.45,7.90]	59.00%	83.00%	73.49%	82.61%
3,3	2,1	[3.34,4.13]	[3.21,3.50]	80.00%	85.00%	77.23%	88.17%
4,5	5,5	[1.11,1.69]	[1.22,1.48]	94.00%	98.00%	89.71%	95.86%
5,0	3,2	[4.27,4.66]	[4.23,4.55]	70.00%	96.00%	79.76%	93.88%
6,0	1,5	[11.34,12.00]	[10.27,10.68]	70.00%	82.00%	59.30%	94.38%
1,2	2,6	[5.73,6.35]	[5.21,5.52]	63.00%	79.00%	71.79%	94.38%
2,5	6,0	[9.57,10.67]	[9.06,9.29]	29.00%	57.00%	71.01%	92.41%
5,1	0,4	[9.16,9.73]	[8.02,8.66]	70.00%	75.00%	67.90%	95.45%
0,1	6,1	[6.61,7.06]	[6.61,6.97]	74.00%	85.00%	74.16%	85.82%
4,0	3,3	[4.23,5.17]	[4.26,4.62]	71.00%	85.00%	88.66%	90.11%

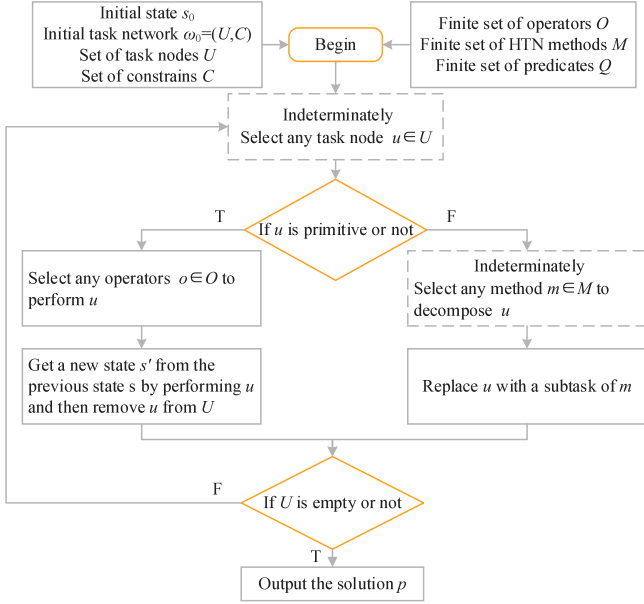


Fig. 16. The schema of HTN approach.

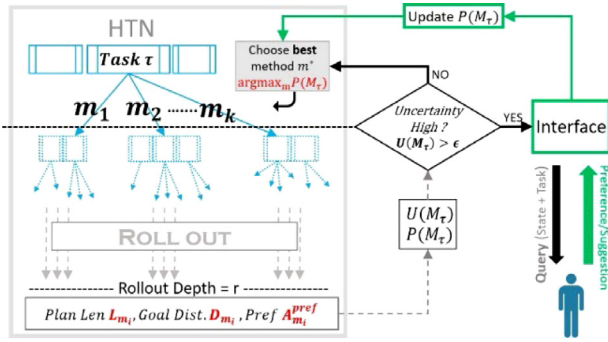


Fig. 17. Overall architecture of PGPlanner (Das et al., 2019).

and curiosity at a high-level hierarchical reinforcement learning. In the long run, discovering both optimal and user-friendly solutions is desirable and glamorous.

4.2. HTN for multi-task

HTN planning has a hierarchy of task decomposition and expressability of domain knowledge representation, which is suitable for large-scale practical planning problems, for example, robot task planning for home service. Robots providing service at home should be able to determine the efficient task sequence that satisfies the users and meets needs of themselves when the users have multi-task requirements. Lu

et al. (2016) built an improved JSHOP2 planner (SHOP2 written in Java) to optimize multi-task planning in home scenes. Qi et al. (2017) introduced resources and temporal constraints to handle multi-capacity discrete resources and complex time constraints simultaneously, though the dynamic environment is not taken into consideration. Afterward, Bercher et al. (2019) and Olz et al. (2021) proposed adding preconditions and effects to compound tasks for motivation that making them directly visible and exploiting them for search. Recently, Cheng et al. (2018), Hoeller et al. (2020b) and Holler et al. (2019) attempted to introduce heuristics to improve the overall performance of HTN planners and even outperform the state-of-the-art HTN planners. However, most of the work have been focused on finding a solution as quickly as possible and very little has been done to solve for optimal plans. Only (Behnke et al., 2019) were devoted to finding optimal solutions for HTN planning. In addition, Hierarchical Domain Definition Language (HDDL), an extension to PDDL for expressing hierarchical planning problems (Hoeller et al., 2020a) was proposed to satisfy the needs of hierarchical planning systems. Besides, Icarte et al. (2018), Lee and Choi (2022), Li et al. (2021) and Takayanagi et al. (2019) introduced learning-based methods to enable robots to plan and take appropriate actions to accomplish complex tasks. When multiple methods are applicable for a compound task, how to select the proper method is shown in Fig. 19.

Takayanagi et al. (2019) focused on learning a policy that can break down a given task hierarchically from the object's goal state and the current observation, which enables robots to plan and take appropriate actions to accomplish complex and unseen tasks. Okumus et al. (2020) developed a centralized cloud system that makes multi-agent work in collaboration and designed a task allocation algorithm to achieve optimal performance of tasks, which may inspire the further development of home service. Planning for multi-task is still in its early stage and many problems remain to be investigated. Some promising approaches are discussed and presented in the final section.

4.3. HTN for re-planning

The dynamic nature of the real environment is likely to lead to the failure of the original planning and thus re-planning is expected. HTN planning is a good method on-demand since it can postpone subtask until it needs to be executed under task decomposition property, which makes starting from scratch avoided, and hence efficiency will be guaranteed.

A method that recommended more detailed planning for short-term actions rather than later in the planning period was proposed (Kaelbling and Lozano-Perez, 2016). Li et al. (2017) developed a heuristic HTN planning method integrating planning generation and execution, which can check and adjust the original planning to lessen unnecessary re-planning in variable execution environments. Because disruptions in the real world will cause adverse results, novel and more effective methods are desired to further reduce unnecessary re-planning. A strategy combining symbols and motion planning was proposed for the manipulation tasks of bimanual manipulation, in which HTN is used to encode different sequences of motion and deal with possible

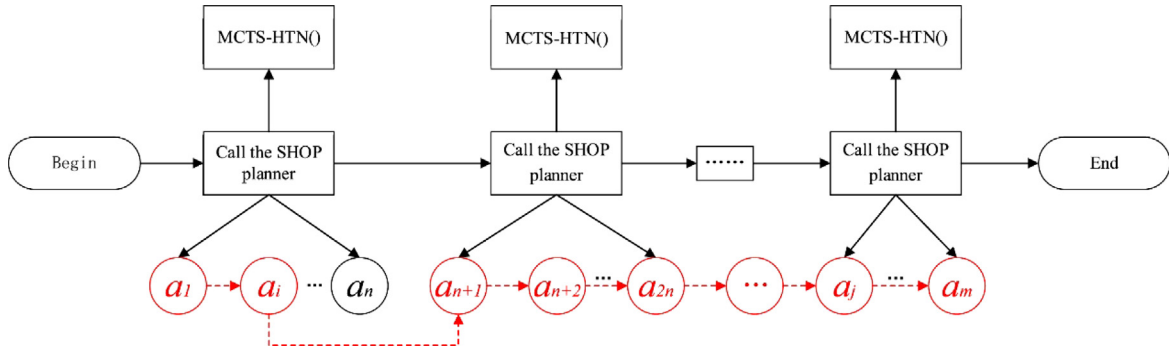


Fig. 18. Planning and executing the process of SHOP-m in a dynamic environment (Shao et al., 2021).

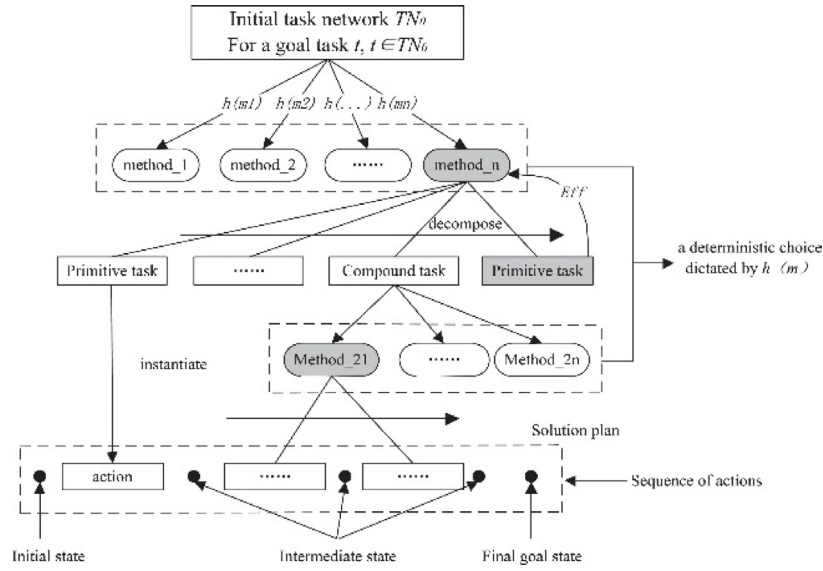


Fig. 19. The method selection process in SHOP-h (Cheng et al., 2018).

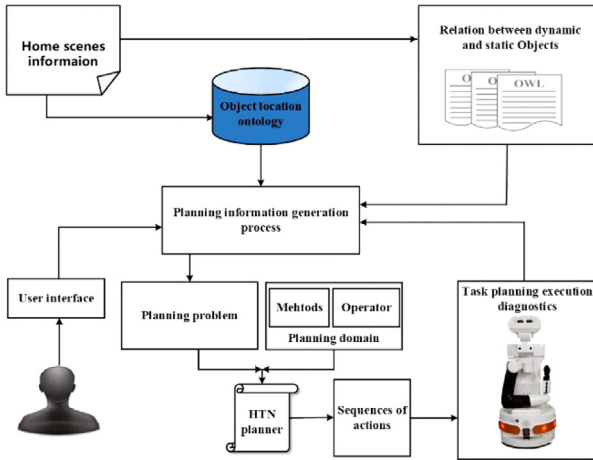


Fig. 20. Diagram of home service robot task planning with HTN planner (Wang et al., 2020).

interferences between dual-arm (Suarez-Hernandez et al., 2018). As a result, sub-goals can be easily identified and are likely to be planned separately for each goal. Meanwhile, the uncertainty of the current state is handled and reduced by observing actions and environmental reshaping actions. However, more efforts should be made to solve dynamic and exogenous issues.

Harman et al. (2019) developed a Hierarchical Continual Planning for delaying the actions that will not be executed immediately and thus the frequency of re-planning is reduced. Wang et al. (2020) introduced a task execution diagnostic and re-planning mechanism to update the initial states of the task planner and regenerate sequences of actions when the target object is not found directly as there are so many dynamic objects in the home environment, which significantly improves the flexibility of planning. The diagram of home service robot task planning with HTN planner (HTNP) is shown in Fig. 20.

Moreover, the task planning time comparison of five typical tasks, i.e., putting the fruit in the refrigerator (T1), sending milk (T2), pouring a cup of water (T3), food heating (T4), and sending a book (T5) with Human-based task planning (HBTP) and comprehensive search-based task planning (CSTP) is illustrated in Fig. 21.

As seen in Fig. 21, the method proposed by Wang et al. achieves better performance than CSTP and is close to the HBTP, which is beneficial to improve the autonomy and effectiveness of the home service robot. Future work should take the initial location of the robots and the task-related objects close to the robots into consideration.

4.4. Task intelligence with hierarchy

Task intelligence is a method of generating task sequences by using hierarchical thought. As the name suggests, adaptability and intelligence are the highlights. Kim et al. (2013) proposed a task intelligence system to achieve general intelligence of robots by mimicking the basic functions of the human brain in robot aspects, which consists of five

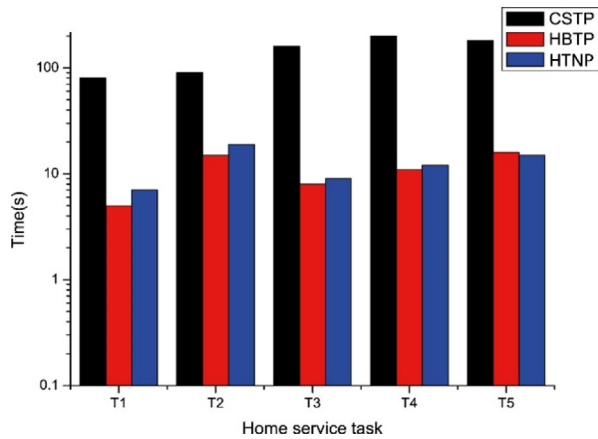


Fig. 21. Task planning time comparison of five typical tasks with different methods (Wang et al., 2020).

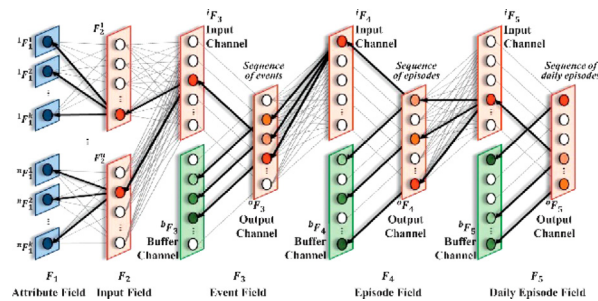


Fig. 22. Deep ART architecture (Kim et al., 2017).

parts, namely, perception collects sensory data separately from the external environment and the internal state of robots, memory stores information through the learning module of the reasoning part and improves the future behaviors of robots, reasoning utilizes contextual information, internal state and memory to determine high-level actions, and internal state and execution parts. In practice, episodic memory plays an important role because it satisfies the requirements of intelligent robots to memorize and retrieve a sequence of events from past experiences, thus providing proper service to users.

To make full use of the cognitive ability of episodic memory, an unsupervised neural network model of episodic memory called Deep Adaptive Resonance Theory (Deep ART) with incremental clustering ability (Park and Kim, 2016) was proposed. The structure of Deep ART is shown in Fig. 22. Fig. 23 illustrates the details for realizing the task intelligence.

Kim et al. (2016) integrated Deep ART into semantic memory, which can reduce the time complexity of solving given problems. However, environmental conditions or states cannot be categorized and defined from the perspective of behavior planning before the running planner obtains the solution. Jeong et al. (2017) utilized Deep ART, a neural model-based mechanism of thought, and a motion planning algorithm to enable robots to have memory, reasoning, and execution capabilities, which are keys to achieve robot task intelligence. Kim et al. (2017) and Llopert et al. (2018) employed Deep-ART, FF-planner (an action sequence generator) (Hoffmann and Nebel, 2001), and an adaptive behavior generation that can generate the modified trajectory without colliding with surrounding obstacles. All of these make it possible for robots to behave in dynamic environments.

Apart from Deep-ART, novel ART theories (Choi et al., 2021; Kim and Kim, 2020; Nasir et al., 2019; Park et al., 2018) were proposed for performing smart service in a changing environment. Table 6 compares these theories with their performance, experiments, limitations, and

possible improvements. It is illustrated that none of these methods is perfect and more rigorous processing is required.

Furthermore, in order to complete family service tasks with the cooperation of others, Lee et al. (2018) employed memory and reasoning to realize planning adaptively and perceived users' behaviors and objects through an RGB-depth and thermal sensor. In the future, solving complex tasks instead of just simple manipulations like putting and pouring, should be focused on.

As a fascinating topic, task intelligence is still in its infancy and significant open challenges and research questions remain. Except for the memory-based method currently getting a lot of attention, learning-based methods which can improve the actions of robots in the future are also important and worth paying attention to.

5. Learning-based methods for task generalization

Learning-based methods with robustness and dynamic adaptability are the core of task planning, as there are plenty of variations in the real world that robots cannot obtain accurate models of environments, objects, or essential skills to manipulate them in advance (Kroemer et al., 2021; Pajarinen and Kyrki, 2017; Riccio et al., 2021). In the learning-based methods, a robot learns the correct sequences of actions by itself (called active learning), a "teacher" (called learning from demonstration), or the existing experience. The research on realizing task planning and task generalization is discussed in consideration of the characteristics and volume of the triple kinds of learning-based methods.

5.1. Active learning

Sometimes, the domain changes so dramatically that robots have to improvise. Delightfully, active learning methods can provide great solutions for discovering new skills amid changes. The term "active learning" was coined in 1994 by Cohn et al. (1994) to describe a learning method that has some control over training inputs. The capacity enables learning ability more powerful and generalization better, which is suitable for real-world scenes.

5.1.1. Active learning dependent on HRI

Moreover, it is necessary to consider planning dependent on HRI due to the variability and uncertainty of environments. Jung et al. (2007) proposed a cognitive case-based reasoning framework (CBR) and introduced the cognitive HRI paradigm to make robots perform tasks at home. However, it can only solve a very limited number of services. Mininger et al. (2018) proposed an interactive task learning method that integrates target tasks and program tasks, which can learn multiple tasks after a single training example with high efficiency. Caccavale and Finzi (2021) and Kim and Yoon (2014) attempted to enhance the adaptability in practical service-providing for humans. Morere et al. (2019) and Wang et al. (2021b) devoted to making robots learn to plan complex and dynamic tasks and showed that active learning can reduce the number of trials required for robots to learn skills. The current methods focus on learning in simulation and transferring successful outcomes to the real world rather than developing approaches that only need to learn from minority real-world samples without simulation. In addition, a robot cannot have an accurate model of environment due to the great number of variations in the real world. More detailed information about how robots represent objects and learn features hierarchically and how to estimate these features and object properties by using passive and interactive perception was explained in Kroemer et al. (2021).

Human action recognition (HAR) based on wearable sensor is very crucial in the process of interacting with humans. The recognition accuracy largely depends on the feature extraction ability of the system. Recurrent neural networks, convolutional neural networks (CNN), and long short-term memory (LSTM) have been proven to be promising

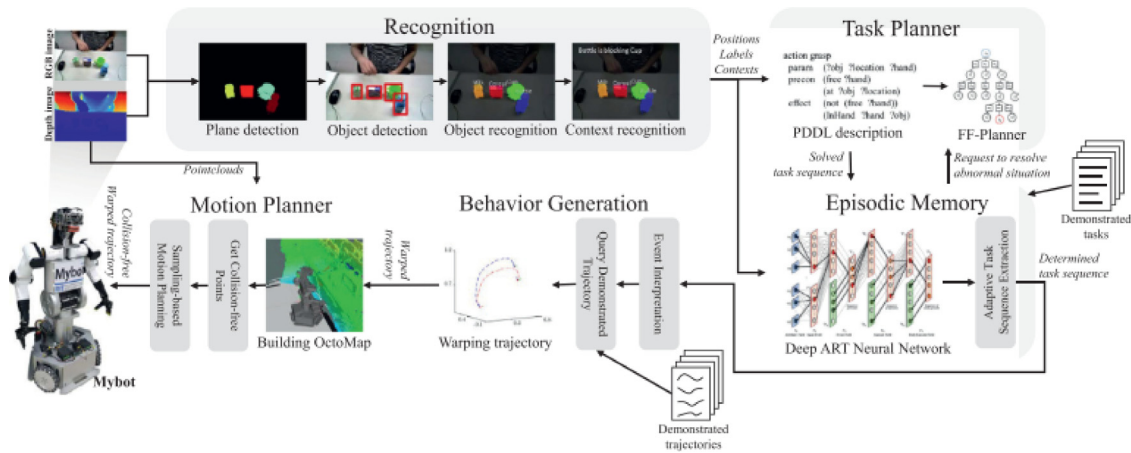


Fig. 23. The system overview to realize the task intelligence (Kim et al., 2017).

Table 6
Comparison of various methods for novel ART theories.

Ref.	Objective	Method(s)/algorithms	Accuracy/performance	Experiments	Limitation/possible improvements
Park et al. (2018)	Providing smart service and undertaking trouble-some or difficult tasks	A novel Deep adaptive resonance theory (ART) to assess task performance	Effectiveness and applicability	Three tasks of arranging toys, making cereal, and disposing of garbage.	Adding contextual information
Nasir et al. (2019)	Automated task planning for robots	User preference-based integrated multi-memory model (pMM-ART)	Enhancing the behavior of an autonomous agent.	Moving apple from counter to fridge	Generalization of semantic similarity, evaluation in a comprehensive dataset
Kim and Kim (2020)	Offering fully customized service to its residents	Stabilized feedback (SF) ART	Learning and reasoning and improving stability and versatility	Providing service to the user in a kitchen environment	Conducting mathematical analysis and finding out the principles behind it
Choi et al. (2021)	Tackling limitations of ART	Semantic relation-aware episodic memory (SR-EM)	Providing home service more intelligently than the one with Deep ART	Serving a coffee can to the table and capturing objects of similar shape	Sim to real

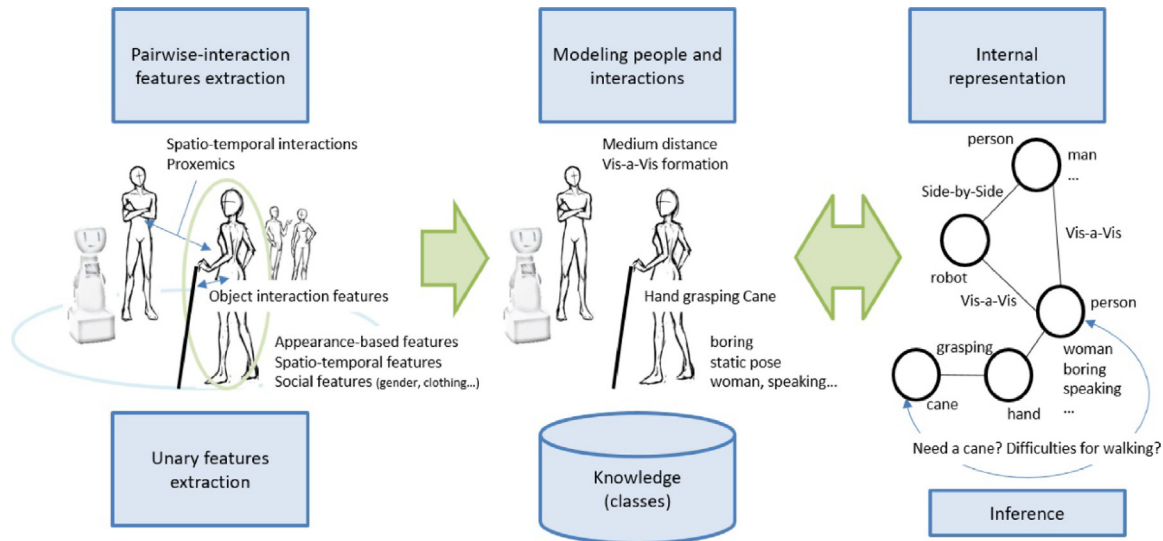


Fig. 24. Major modules on a system in charge of modeling human behaviors and interactions (Tapus et al., 2019).

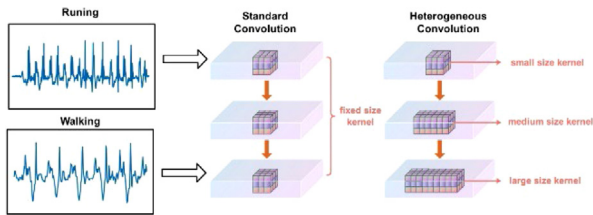
techniques in this field (Tapus et al., 2019). The main modules of a system responsible for modeling human behaviors and interactions are shown in Fig. 24, which typically include feature extraction and classification, internal representation, and inference mechanisms. But the pipeline plan is only partially real. Internal representations can store raw data from feature extractors and help model the entire scene.

Cheng et al. (2022), Han et al. (2022) and Xiao et al. (2021) obtained excellent recognition results on the public benchmark HAR datasets by improving CNN-based methods. Tang et al. (2022) and Xiao et al. (2021) introduced an attention mechanism for mining relationships hidden in data. Meanwhile, Chen and Shi (2021), Xing et al. (2022a) and Xing et al. (2022b) focused on the realization of feature extractors for time series classification (TSC). More information about

Table 7

Comparison of various algorithms for feature extraction.

Ref.	Objective	Method(s)/algorithms	Datasets	Accuracy/performance	Limitation/possible improvements
Xiao et al. (2021)	Wearable sensor-based HAR	Convolutional block, LSTM, and attention mechanism	WISDM, UCI-HAR 2012, OPPORTUNITY, and PAMAP2	Outperforming 14 existing HAR algorithms and can be applied to real-world	Improving recognition accuracy
Cheng et al. (2022)	Real-time HAR using mobile and wearable sensors	Conditionally parametrized convolutions	WISDM, PAMAP, UNIMIB-SHAR, OPPORTUNITY	Replacing the existing deep HAR architecture and can be applied to real-world	Decreasing computation cost
Han et al. (2022)	Wearable sensor-based HAR	Heterogeneous CNNs	OPPORTUNITY, PAMAP2, UCI-HAR, USC-HAD, the Weakly Labeled HAR	Easy to be integrated into standard convolutional layers	Better tradeoff privacy protection and recognition accuracy at different time scales
Tang et al. (2022)	Wearable sensor-based HAR	Triple Cross-Domain Attention	UCI-HAR, PAMAP2, WISDM, and UNIMIB-SHAR	Good generality ability and visualization analysis	Applying to practical HAR scenarios
Xu et al. (2022)	Wearable sensor-based Multimodal HAR	Deformable Convolutional Networks	OPPORTUNITY, UNIMIB-SHAR, USC-HAD, and WISDM	State-of-the-art recognition accuracies, e.g., 82.91%, 80.02%, 97.35%, and 99.21%	Leveraging intermodulation information to connect the deformable convolution and attention modules
Chen and Shi (2021)	STC	Multi-scale Attention Convolutional Neural Network	UCR standard datasets	The best performance and outperforming the other traditional and deep learning-based methods	Exploring ensemble methods of different deep learning architectures to achieve better performance
Xing et al. (2022a)	Multitask TSC	Federated distillation learning system (EFDLS)	UCR2018 standard datasets	Applying to multitask TSC problems in various real-world domains	Validating EFDLS on more real-world datasets
Xing et al. (2022b)	STC	The semisupervised model with self-distillation (SD)	UCR2018 standard datasets	Applicable to TSC problems in the real world	Properly setting many hyperparameters to obtain the optimal performance

**Fig. 25.** The comparison between normal convolution and heterogeneous convolution scaling for different activities (Han et al., 2022).

the HAR can be found in Tapus et al. (2019) and Zhang et al. (2022a). Two architectures of CNN are shown in Figs. 25 and 26, respectively. Table 7 compares various algorithms for feature extraction with their objective, performance, limitations, and possible improvements.

5.1.2. Task generalization of active learning

Generalization focuses on discovering the inherent laws of the task to understand and reason for complex tasks and better plan fresh tasks, which plays an important role in achieving intelligent task planning. Many scholars have also been attracted to task generalization and made achievements. Masson et al. (2016) first introduced and studied the concept of parametric action space Markov decision-making process, in which discrete action with continuous parameters constitutes a parametric action space to provide adaptability of continuous domains and use distinct kinds of actions. Andreas et al. (2017) and Yahya et al. (2017) focused on using policy to achieve task generalization but the former lacks manipulation with a robotic arm and the latter does not have a mobile base.

Based on self-guided reinforcement learning (RL) and imitation learning, Zhu et al. (2017) developed a deep predictive model based on successor representations to ensure cross-task generalization. However, transferring knowledge to the real world and realizing more complicated tasks with a richer set of actions should be further developed. Jing et al. (2019) migrated the action policy to the target task from the source with preference-based cost learning, which is the goal of task transfer in RL. Fang et al. (2021) introduced Skill Learning in

Diversified Environments to discover generalizable skills by learning high diversity between and within skills. Moreover, an RL policy in the hierarchy was designed and combined with discovered skills, which can effectively improve the robot's performance when target tasks are invisible. Utilizing procedural content generation to realize robot learning is expected.

A condition-generation method for converting text manipulation instructions into a sequence of actions was proposed to automate knowledge-providing for robots and guide the implementation of unfamiliar homes (Zhang et al., 2021). Moreover, Riccio et al. (2021) introduced an iterative learning algorithm, LoOP that integrates MC search and Q-learning to generate action strategies, which overcomes the lack of generalization in task planning between similar states.

Table 8 compares the various methods for task generalization of active learning with their performance, experiments, limitations, and possible improvements. It is illustrated that RL is the key to realizing task generalization of active learning. The process of implementing RL is illustrated in Fig. 27, in which the agent interacts with the environment by selecting actions with the policy and achieves the goal through optimization under the interaction.

Transfer is also a promising approach for achieving the performance of target task generalization. Kandaswamy et al. (2017) developed deep transfer learning among multiple domains of information by extending the version of the source–target–source. Ding (2020) studied the skill transfer from human arms to anthropomorphic ones, which can make robots adapt to various tasks flexibly. Zhang et al. (2022b) proposed a transfer learning method for multi-task based on the kernel representation to improve the performance of the target learner. Lack of interpretability, however, task generalization will confuse users and its intelligence cannot meet expectations.

5.2. Learning from demonstration

Except for exploiting active learning to acquire new skills, learning from demonstration (LfD) is also a convincing way while ideal actions are neither easily programmed nor defined as an optimization problem, but can be demonstrated (Ravichandar et al., 2020). Most importantly, experts are not always required, and the flexibility and compliance of humans can be well-mapped to robots, which will greatly facilitate the task completion of robots at home (Yang et al., 2019).

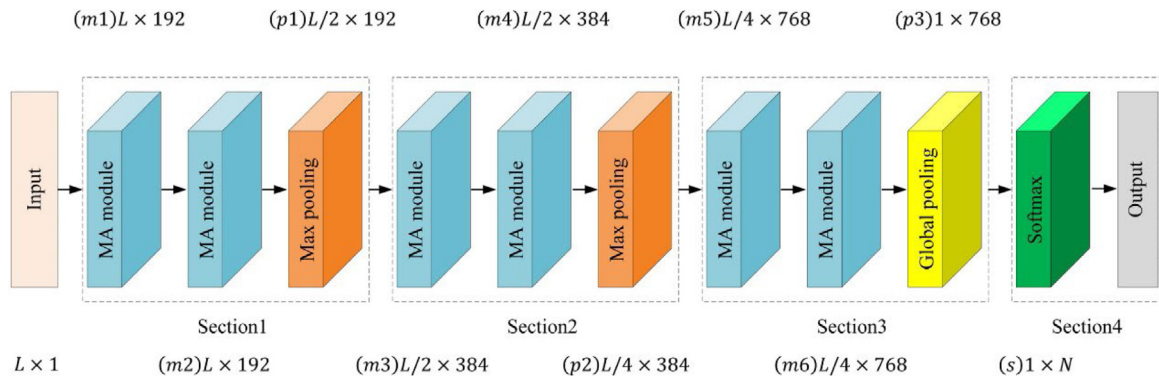


Fig. 26. Overall architecture of Multi-scale Attention Convolutional Neural Network (Chen and Shi, 2021).

Table 8

Comparison of various methods for task generalization of active learning.

Ref.	Objective	Method(s)/algorithms	Accuracy/performance	Experiments	Limitation/possible improvements
Andreas et al. (2017)	Sharing behaviors across tasks	Multitask deep reinforcement learning guided by policy sketches	Library of interpretable primitive behaviors that can be recombined to rapidly adapt to new tasks	Avoiding obstacles and interacting with various kinds of objects in three types of environment	No manipulation with a robotic arm
Yahya et al. (2017)	Achieve generalization	Reinforcement learning distributed and asynchronous policy learning	Improving training times on challenging and real-world manipulation tasks	Multiple robots collaborate to learn the door-opening skill	No mobile base
Zhu et al. (2017)	Cross task generalization	A deep predictive model based on successor representations	Near-optimal results	Interacting with the scene and predicting a sequence of actions to achieve the goal based on visual inputs	Transferring knowledge to real-world and realizing more complicated tasks with a richer set of actions
Jing et al. (2019)	Task transfer	Preference-Based Cost Learning in RL	Theoretical analysis of the distribution learning and convergence	Several control tasks in MuJoCo	Quantitative evaluation of the improvement on the transferred cost function
Fang et al. (2021)	Generalizable skills	Skill Learning In Diversified Environments, hierarchical reinforcement learning	Improving the robot's performance in various unseen target tasks	Two tabletop manipulation domains	Procedural content generation for robot learning
Zhang et al. (2021)	Guiding robots to perform unfamiliar home tasks	RL, a two-phase conditional generation method integrating a priori knowledge, semantic similarity, and action logic	Similarity evaluation	Cleaning a table in a real scene	Further studying the relationship between objects and location labels
Riccio et al. (2021)	Generalization among similar states	LoOP: iterative learning combination of Monte-Carlo Search Planning and Q-learning	Robustness and flexibility in various domains and multiple robotic platforms	Fetching task	Extending formulation to learn policies in the continuous action space

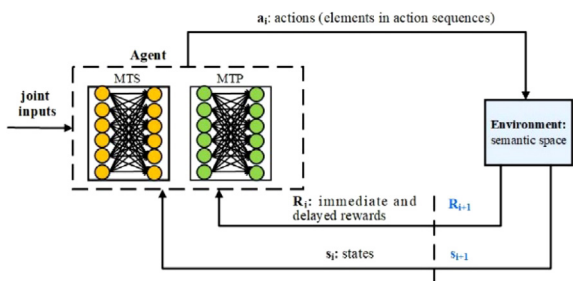


Fig. 27. The process of implementing reinforcement learning (Zhang et al., 2021).

5.2.1. Few demonstrations

Few demonstrations are expected especially when “teachers” are neither experienced nor expert users, but ambiguity is likely to arise. Several studies have been done to disambiguate. Welschhold et al. (2019) combined learning action models with a probabilistic planning framework based on MTSC for computing feasible sequences of actions that mimic the teacher’s intention under the new setting without specifying the target state explicitly. However, the generalization

performance needs to be enhanced. Pignat et al. (2022) proposed an expert-product-based framework to encode distributions in robots so that robots can learn from a few experiments or demonstrations. Chisari et al. (2022) introduced corrective and evaluative feedback from humans for robotic manipulation to tackle real-world learning problems. Nevertheless, human intervention will hinder the progress of robot autonomy, there are some works devoted to making robots fully automated. Carpio et al. (2019) presented a Temporal Reasoning-based LfD framework for learning and automating sequential HRI, even after a small amount of demonstration training. Eteke et al. (2021) introduced a black box specifically policy search (PS) approach with an adaptive step to make convergence rapid. Since demonstrations from non-experts are not always guaranteed to be optimal, determining how to let robots learn from both successful and failed demonstrations is an important issue. Hertel and Ahmadzadeh (2021a) proposed a novel LfD representation to solve this problem by encoding and constructing a set of quadratic costs to find an optimal reproduction. Promising future work includes incorporating techniques such as skill improvement into the learning process, as well as more sophisticated temporal reasoning devices to make the learning process more efficient, fully automated, and more generalizable.

5.2.2. Task generalization of LfD

Although LfD has proven to be a promising means of programming quickly and naturally, it is often limited because of poor generalization. Many scholars have carried out relevant research to strengthen generalization encouragingly. Niekum et al. (2015) provided decisive insight into characteristics that are task importance, task invariants, advanced task structure, and appropriate skills by automatically detecting and exploiting repeated structures of multiple levels of abstraction in demo data, which facilitates the representation of finite-state and robust generalization as well as the transfer of complexly multi-step tasks. The difficulties to make robots enter home will not be limited to subfields such as perception or control, but a combination of them. Dianov et al. (2016) and Hertel and Ahmadzadeh (2021b) improved skill generalization by exploiting the similarities between task parameters. Furthermore, the similarity between different task structures should be explored in more detail. Zhang et al. (2018) used Bayesian inference and Gaussian Mixture Model to split complex tasks into movement primitives (MPs) for learning complex sequential tasks, but the method is time-consuming as it must learn every segment from scratch. Huang et al. (2019) parsed individual demos into programs, which are then transferred to robot execution in a new scene. Yang et al. (2019) aimed at promoting the generalization of robotic skills and proposed a new framework for skills segmentation, encoding, and adjusting. Moreover, thanks to the model-free model, it can be adapted to other platforms. Integrating RL techniques will improve policy effectively.

Although parameterized skill representation allows generalization in new scenes, it is limited in scope and slow. To solve this, Le et al. (2021) extended the LfD framework to address manipulation skills involving forces; Guo and Burger (2021) proposed a hierarchical and combinatorial planning framework that can learn geometric task networks from detailed planners without any human input while utilizing task parameter models to achieve significant improvements in the efficiency of offline learning, online performance and transparency of the decision-making process.

Moreover, a promising prospective meta-learning approach was proposed that makes robots learn from demonstrations with the idea of “learning to learn” for the purpose that robots can perform as wide-ranging tasks as humans. However, most meta-learning methods focus only on learning from a single domain of demonstration. Considering that humans can acquire and incorporate knowledge from a variety of related fields, Hu et al. (2022) proposed an efficient Random Domain-Adaptive Meta-Learning (RDAML) framework that enables robots to learn from multiple domains of demonstrations with different random parameters, such as humans and robotics. The sources of learning will be further expanded to teach robots to solve more complex cross-domain tasks in the future.

Since robots are prone to anomalies when performing manipulation tasks in an unstructured environment, robots are required to adapt to changes in the environment quickly. Wu et al. (2021) presented a framework for robots to learn recovery skills and variability from human demonstrations and reproductive uncertainty through given safe passage points. As can be seen in Fig. 28, mean-prior Gaussian process regression (MP-GPR, above) and kernelized MPs (bottom) learn the effective performance of the robot’s abnormal recovery skills based on five time-driven trajectories, but the modulation trajectory of probabilistic MPs (medium) cannot avoid obstacles, which indicates that the proposed MP-GPR method allows the robot to avoid obstacles by adjusting the trajectory to a safe point, and easily adapts to environmental changes, such as different target positions and the shape of obstacles. Table 9 compares the various methods for task generalization of LfD with their performance, experiments, limitations, and possible improvements.

While LfD can significantly speed up the transfer of knowledge to robots in unconstrained actions such as picking and placing, the transfer of contact- or force-based skills to robots remains an open question.

Besides, intelligent robots in the home can also enhance their competence from accumulative experience through continuous interaction with the environment, especially with humans (Chen et al., 2018; Lim et al., 2014; Mokhtari et al., 2014). In addition, Mokhtari et al. (2017) and Mokhtari et al. (2019) conducted a series of studies on experience-based planning domains (EBPDs). Kim et al. (2022) presented design principles to learn from experience and representations for effective generalization, in which both plan and data efficiency are improved. Few experience-based learning methods have been retrieved, the discussion here intends to attract further research on this issue.

6. Discussions

Approaches that facilitate adaptive and intelligent task planning have been broadly investigated. Probabilistic methods with online planning enable reasoning and planning in the context of uncertain and incomplete information and to proactively adapt to changes, and the knowledge of home service can eliminate uncertainty is also discussed. Furthermore, the HTN method, which can well reflect the way that users treat problems, is concerned. Its achievements in HRI, multi-task planning, and re-planning are commented on. Particularly, the latest development of task intelligence based on hierarchical thinking is introduced to provide new solutions. Further discussion shows that the methods based on active learning and LfD paint a prospective blueprint, they are robust and adaptable to deal with dynamic changes and can realize task generalization. However, key problems are still up in the air, such as the customizability, interpretability, and predictability of task planning. To further improve the popularity and working ability of robots, and provide a more natural interactive experience and personalized service in the home environment, the following questions can be put forward.

(I) How to overcome the uncertainty of the home environment and generate more reliable, efficient, and scalable task planning? (II) How to make task planning more understandable for users and enhance the predictability of task planning? (III) How to improve learning skills and generalization properties to tackle complex and changeable tasks? So far, these questions have only been partially answered, leaving great chances for further exploration.

For the first problem, solutions can be proposed from the aspects of complementarity of perception, the perfection of knowledge, and the adaptability of different scenes.

(i) Multi-Modal Sensing and Fusion. In addition to the most commonly used visual perception, voice interactions designed to communicate with users and better understand their needs should also be introduced. Meanwhile, the sense of tactile or proximity is a bright spot to achieve safe physical contact and manipulation if the region is visually blocked. Multi-modal perception, obviously, will reduce or even eliminate uncertainty benefiting from the complementarity of multi-modal information. However, the raw data from abundant visual perception, vague commands, and contact touch are extremely diverse and complex. Thus multi-modal fusion methods are required. More discussions on multi-modal fusion methods can be found in Xue et al. (2020).

(ii) Extending the Knowledge Base. It can be seen from the discussion in the previous sections that knowledge plays an important role in task planning. However, knowledge is infinite, the reference objects in the verbal commands from a user are likely not currently represented in its knowledge base, thus expanding knowledge is a top priority. Utilizing open knowledge extracted from the web instead of hand-coding is a bright way (Chen et al., 2012; Lu and Chen, 2017). The anthropomorphic manipulation and skill transfer theory proposed by Ding (2020) provides a strong guide for the realization of LfD and the establishment of the manipulation knowledge base in HRI.

Moreover, knowledge can be transferred among various robot systems, that is, different robots share important information to promote each other and to enhance performance through the Internet of Things

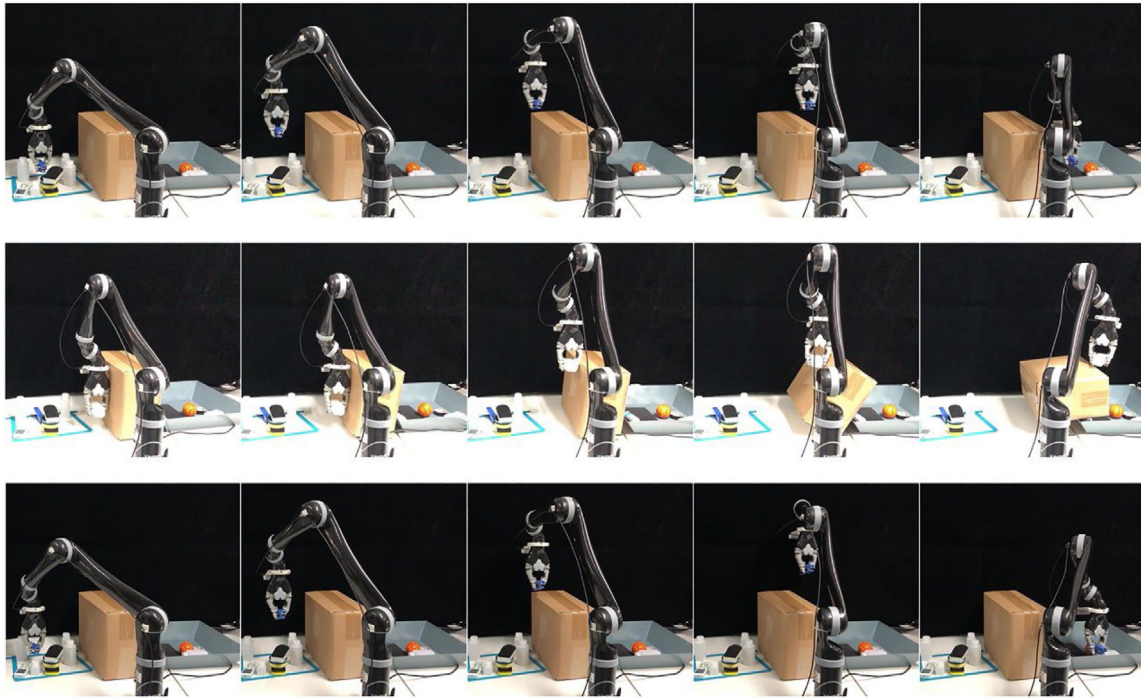


Fig. 28. Different trajectory modulation using MP-GPR, probabilistic MPs, and kernelized MPs under the same via-point configuration (Wu et al., 2021).

Table 9

Comparison of various methods for task generalization of LfD.

Ref.	Objective	Method(s)/algorithms	Accuracy/performance	Experiments	Limitation/possible improvements
Huang et al. (2019)	Complex manipulation capabilities	Parsing a single demonstration into a program and later transferring it to a robot in a new scene for execution	Expressivity and generalizability are verified through an evaluation of a benchmark of complex tasks	Object localization and registration, and grasping the object(s)	Autonomy alongside human supervision
Yang et al. (2019)	Learning and generalizing human-like variable impedance skills	Skills segmentation, encoding, and adjusting	Can be applied to other robotic platforms thanks to the model-free models	Water-lifting task	Integrating RL techniques
Guo and Burger (2021)	Efficient and explainable skill coordination	Parameterized skill and geometric task network (GTN)	The offline learning efficiency, online performance, and transparency of the decision process are improved	Solving various manipulation tasks with a 7-DoF robot arm	Structural composition of GTNs and the online choice of pre-trained GTNs
Hu et al. (2022)	Performing a wide range of tasks like humans	Meta-learning, Random Domain-Adaptive Meta-Learning framework	Better generalization ability and adapting to both human demos and robot demos well	Placing object	Cross-domain tasks learning by learning from humans and robots
Wu et al. (2021)	Rapidly adapting to environmental changes	MP-GPR	Efficiently encode the variability from multiple demonstrations	Bin picking problem	MP-GPR based on visual perception

(IoT) on the condition that security and privacy (Khan et al., 2022; Yang et al., 2022) concerns have been appreciated and solved

(iii) Scalability Realization. Scalability means that models should be able to describe large- and small-scale environments and gradually increase coverage, which has important implications for robots to construct environment models (Rusu, 2010) and perform different tasks while staying in the environments. 3-D scene construction based on semantic information with intuitive usability (Kim et al., 2020; Zheng et al., 2022) and the versatile simulation framework V-REP (Rohmer et al., 2013) are powerful tools to realize broad scalability.

What's more, the realization of customization, explanation, and prediction will greatly improve the satisfaction and comfort of users.

(i) Task/Service Customization. Robots at home as good partners for humans to improve the quality of human life are required to provide customized service for different users, in this case, actively remembering what users prefer and are not satisfied with through a

feedback or evaluation mechanism is a promising means (Kim and Kim, 2020).

(ii) Explanations for Actions. The explanation should be attached of great importance because humans have a strong preference for explanation (Lombrozo, 2006) and the trust of humans can be enhanced by explaining the actions of robots (Edmonds et al., 2019). In addition, the general results produced by the explain-based approach can be transferred to the new variations of a task (Mininger et al., 2018). The ability to provide explanations for the planned sequence of actions is a sign of intelligence, which makes it universally acceptable for robots to perform critical tasks at home.

(iii) Understanding and Prediction. Although online planning or hierarchical methods can be used to re-plan in the face of failure currently, the best is to perceive or predict changes in advance to accommodate unexpected human actions and enable safe interactions in the situation of HRI, which requires that robots must have a better

understanding of how human interact and can adapt in real-time. Furthermore, the popularity of robots serving at home will be greatly increased if robots can predict users' needs according to their preference or habits and provide service proactively. Vision integrating with AI technology (Chu et al., 2018; Lin et al., 2017; Paxton et al., 2019) can provide a reference for achieving this goal.

Last but not least, multi-task learning and human-inspired method are emphasized to improve learning skills and generalization properties.

(i) **Multi-Task Learning.** Multitask learning (Caruana, 1997) is an inductive transfer mechanism that groups multiple related tasks together and trains them in parallel, in which share representation is used to alleviate the sparse problem of task data and avoid overfitting of individual tasks, multi-task learning in parallel can improve the learning efficiency with a larger learning rate, and noise in different tasks will be canceled to some extent for more robust performance. However, multi-task networks are difficult to train due to the critical challenges of sharing and balancing tasks. Casting multi-task learning as multi-objective optimization (Sener and Koltun, 2018) and Multi-Task Attention Network (Liu et al., 2019) provide a beacon to solve this problem.

(ii) **Human-Inspired Method.** Anthropomorphic movement (Ding and Fang, 2013; Fang et al., 2019; He et al., 2021) has enabled robots to achieve human-like motion and guarantee friendly interaction. Similarly, human thought and wisdom about complex task planning and drawing inferences about other cases from one instance based on cumulative experience can also be transferred to robots, which can increase the ability of robots to acquire skills quickly and solve new tasks.

(iii) **Semi-Supervised and Self-Supervised Learning Methods.** To realize perception and recognition in complex environments and robust robotic manipulation for a large number of unknown objects, large-scale training data with annotations is highly required. However, it is super time-consuming and expensive. Semi-supervised (Li, 2020; Zhou et al., 2022) or self-supervised (Nava et al., 2021; Peng et al., 2021) learning methods can reduce the time cost of data acquisition and labeling and bring about significant performance improvements.

Besides, hybrid task planning that integrates multiple task planning methods to foster strengths and circumvent weaknesses may be another fascinating research hotspot for solving complicated task planning problems.

Overall, this study elaborates on various aspects of robot task planning for home service from an adaptive and intelligent perspective, providing researchers working in this field with comprehensive information on all available latest technologies

CRediT authorship contribution statement

Haizhen Li: Conceptualization, Investigation, Writing – original draft, Writing – review & editing. **Xilun Ding:** Conceptualization, Review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix

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	Frank, C., 2017. Modern Robotics-Mechanics, Planning, and Control: Cambridge University Press.
	J. Craig, J., 2016. Introduction To Robotics Mechanics and Control Fourth Edition.
	Siliciano, B., Sciavicco, L., Villani, L., & Oriolo, G., 2010. Robotics: modeling, planning and control. New YorN, NY, USA: Springer.
Book on planning	Spong, M. W., Hutchinson, S., & Vidyasagar, M., 2020. Robot modeling and control: John Wiley & Sons.
	Thrun, S., Burgard, W., & Fox, D., 2006. Probabilistic Robotics: Cambridge, Mass: MIT Press.
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	Stuart Russell, P. N., 2010. Artificial Intelligence A Modern Approach Third Edition: Prentice Hall.
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